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Inventor(s)	Ou; Li et al.

Adaptive system probe action to minimize input/output dirty data transfers

Abstract

Adaptive system probe action to minimize input/output dirty data transfers is described. In one or more implementations, a system includes a processor, a memory configured to store data, and a cache configured to store a portion of the data stored in the memory for execution by the processor. The system also includes a cache coherence controller including a cache line history. The cache coherence controller is configured detect a direct memory access request from an input/output device. The direct memory access request is associated with an input/output operation involving the data. The cache coherence controller is further configured to identify a cache line associated with the direct memory access request, and, in response to the cache line history including a dirty data transfer record corresponding to the cache line, selectively send a probe to the cache based on a state of the cache line.

Inventors: Ou; Li (Austin, TX), Balakrishnan; Ganesh (Austin, TX), Apte; Amit (Austin, TX)

Applicant: Advanced Micro Devices, Inc. (Santa Clara, CA)

Family ID: 1000008820067

Assignee: Advanced Micro Devices, Inc. (Santa Clara, CA)

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Primary Examiner: Rigol; Yaima

Attorney, Agent or Firm: FIG. 1 Patents

Background/Summary

BACKGROUND

(1) Various operations of computing systems involve transferring data between memory (e.g., main memory) and other devices (e.g., peripheral devices). Advances in memory access techniques continue to be sought to improve performance of the computing systems. One example advance is direct memory access (DMA), which is a technique used to enhance data transfer efficiency between devices and memory. Instead of relying on a central processing unit (CPU) for every data transfer, DMA allows the devices to directly access the memory without CPU intervention. This greatly reduces the CPU's workload and improves overall system performance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram depicting a non-limiting example system having a system-on-chip (SoC), a memory, and one or more I/O devices.
- (2) FIG. 2 is a block diagram depicting a non-limiting example of a cache coherence controller and a cache.
- (3) FIG. 3 is a block diagram depicting non-limiting example interactions among various components of the system depicted in FIG. 1.
- (4) FIG. 4 is a flow diagram depicting a procedure in a non-limiting example implementation of a DMA controller configured to perform operations to minimize I/O dirty data transfers.
- (5) FIG. 5 is a flow diagram depicting a procedure in a non-limiting example implementation of a cache coherence controller configured to selectively execute an adaptive algorithm.
- (6) FIG. 6 is a flow diagram depicting another procedure in a non-limiting example implementation of a cache coherence controller configured to perform operations to minimize I/O dirty data transfers.

DETAILED DESCRIPTION

Overview

(7) In I/O workloads, where there are substantial interactions between cores and I/O devices, the complexity of such engagements can be reduced to a simpler producer-consumer model. In this model, the producer generates I/O data, and the consumer processes it, with DMA buffers, allocated into system memories, serving as the primary interfaces for their interaction. For instance, in an application where a high-speed network interface card (NIC) is used to transmit data, the cores act as producers, preparing and writing network packets into the DMA buffers, while the NIC, as the consumer, reads the packets from the DMA buffers, transmitting the packets to the network.

(8) Within this model of interaction, high-performance I/O devices necessitate the collaborative operation of multiple cores to produce or consume I/O data sufficiently rapidly to support the high I/O speeds that modern I/O devices can deliver. In a standard I/O software stack, the same DMA buffers are shared among multiple cores driving the same I/O device. Taking a transfer control protocol/Internet protocol (TCP/IP) implementation as an example, a shared buffer pool is pre-allocated for each compute non-uniform memory access or “NUMA” node. All cores from the same compute NUMA node retrieve buffers from a common pool before driving the NIC to transmit data, and subsequently return the buffers to the pool after the NIC has completed its transmission, leading to potential issues with the high latency of the transfer of dirty data among cores, and possible impacts on performance and I/O device capability.

(9) Under current coherence protocols, specific operational considerations come to light in the handling of dirty data within the context of DMA read requests. A conventional procedure involves dispatching probes from a coherence point to caches, the objective being to fetch dirty data without inducing a state change in a cache line—a no-operation (no-op) action. Upon the production and subsequent consumption of I/O data by a core and I/O device respectively, through the DMA reads, related DMA buffers are reintegrated into the buffer pool, being rendered available for impending I/O transactions. Post-consumption, modified cache lines residing in the original core cache are rendered functionally redundant, indicating that their retention does not directly serve subsequent computational or transactional processes.

(10) The systems and techniques disclosed herein change the probe action sent to core caches responsive to DMA read requests. Instead of proceeding with normal operations (i.e., not changing the cache line state), the coherence point functions in one of two probe modes for DMA reads according to historical records that identify whether dirty data transfers were triggered by previous requests for the same cache line. In a first mode, the coherence point functions normally and does not change the cache line state. In a second mode, the coherence point changes the cache line state

to avoid future dirty data transfers. In one or more implementations of the second mode, the coherence point changes the cache line state either by invalidating the cache line or by converting the dirty cache line to a clean cache line.

(11) In one or more implementations, the coherence point is a cache coherence controller implemented in hardware, software, or a combination of both hardware and software. The cache coherence controller is configured to execute an adaptive algorithm to perform operations such as those described above. To summarize briefly, the adaptive algorithm enables the cache coherence controller to minimize unnecessary later dirty data transfers by switching a probe action sent to core caches on-the-fly according to system data sharing patterns. The adaptive algorithm forces cache line invalidation or converts cache line state from dirty to clean when dirty data transfers were observed from a cache line. The adaptive algorithm also switches to no operation probes if data sharing is minimized.

(12) The systems and techniques described herein provide several advantages over existing solutions. For example, the disclosed systems and techniques deliver high I/O performance to customers under default or out-of-the-box configurations without complex software modifications. This is because the adaptive algorithm is implemented as part of the cache coherence controller, which is included as part of a processor package (e.g., a system-on-a-chip) and is not reliant on customer-side software configurations for implementation. Moreover, this is particularly valuable because end users are often reluctant to implement complex software changes that divert from default software configurations. Moreover, in many use cases, customized software configurations are not viable in a customer's production environment. The systems and techniques described herein provide at least these advantages over existing solutions.

(13) In some aspects, the techniques described herein relate to a system including a processor, a memory configured to store data, a cache configured to store a portion of the data stored in the memory for execution by the processor, and a cache coherence controller including a cache line history, the cache coherence controller configured to identify a cache line associated with a direct memory access request, the direct memory access request associated with an operation involving the data, and responsive to the cache line history including a dirty data transfer record corresponding to the cache line, selectively send a probe to the cache.

(14) In some aspects, the techniques described herein relate to a system, wherein the cache coherence controller is configured to selectively send the probe to the cache based on a state of the cache line, the state of the cache line including a modified state, an exclusive state, a shared state, or an invalid state.

(15) In some aspects, the techniques described herein relate to a system, wherein the cache coherence controller is further configured to, in response to determining that the state of the cache line is the exclusive state, send the probe to the cache to invalidate the cache line.

(16) In some aspects, the techniques described herein relate to a system, wherein the cache coherence controller is further configured to, in response to determining that the state of the cache line is the shared state, send the probe to convert the state of the cache line from a dirty shared state to a clean shared state.

(17) In some aspects, the techniques described herein relate to a system, wherein the processor includes a first core and a second core, and wherein the cache includes a first cache corresponding to the first core and a second cache corresponding to the second core.

(18) In some aspects, the techniques described herein relate to a system, wherein the first cache stores the portion of the data in the cache line.

(19) In some aspects, the techniques described herein relate to a system, wherein the first core transfers the portion of the data from the first cache to the second cache.

(20) In some aspects, the techniques described herein relate to a system, wherein the first core modifies the portion of the data to create a new portion of the data that is different from the portion of the data.

- (21) In some aspects, the techniques described herein relate to a system, wherein the cache coherence controller is further configured to update the cache line history to reflect that the portion of the data was modified before being transferred to the second cache.
- (22) In some aspects, the techniques described herein relate to a system, wherein the operation involving the data includes an input or output operation.
- (23) In some aspects, the techniques described herein relate to a cache coherence controller including a memory configured to store a cache line history, and a hardware circuitry configured to identify a cache line associated with a direct memory access request, the direct memory access request associated with an operation involving data stored in a cache, and responsive to the cache line history including a dirty data transfer record corresponding to the cache line, selectively send a probe to the cache.
- (24) In some aspects, the techniques described herein relate to a cache coherence controller, wherein hardware circuitry includes the memory.
- (25) In some aspects, the techniques described herein relate to a cache coherence controller, wherein the hardware circuitry is configured to selectively send the probe to the cache based on a state, the state including a modified state, an exclusive state, a shared state, or an invalid state.
- (26) In some aspects, the techniques described herein relate to a cache coherence controller, wherein the hardware circuitry is further configured to, in response to determining that the state of the cache line is the shared state, send the probe to convert the state of the cache line from a dirty shared state to a clean shared state.
- (27) In some aspects, the techniques described herein relate to a cache coherence controller, wherein the hardware circuitry is further configured to update the cache line history to reflect that a portion of the data was modified before being transferred to the cache.
- (28) In some aspects, the techniques described herein relate to a cache coherence controller, wherein the operation involving the data includes an input operation or an output operation.
- (29) In some aspects, the techniques described herein relate to a method including detecting, by a cache coherence controller, a direct memory access request from a direct memory access engine of an input/output device, the direct memory access request associated with an input/output operation performed by the input/output device, and responsive to detecting the direct memory access request, selectively executing, by the cache coherence controller, an adaptive algorithm to ensure coherency of data between a plurality of caches and a memory, the data associated with the input/output operation.
- (30) In some aspects, the techniques described herein relate to a method, wherein selectively executing, by the cache coherence controller, the adaptive algorithm includes executing, by the cache coherence controller, the adaptive algorithm in response to the data being associated with a dirty data transfer from a first cache of the plurality of caches to a second cache of the plurality of caches.
- (31) In some aspects, the techniques described herein relate to a method, wherein executing, by the cache coherence controller, the adaptive algorithm includes determining if a state of a cache line of the first cache is an exclusive state or a shared state.
- (32) In some aspects, the techniques described herein relate to a method, further including in response to determining that the state of the cache line of the first cache is the exclusive state, sending a probe to invalidate the cache line of the first cache, or in response to determining that the state of the cache line of the first cache is the shared state, sending a probe to convert the state of the cache line of the first cache from a dirty shared state to a clean shared state.
- (33) FIG. 1 is a block diagram of a non-limiting example system **100** having a system-on-chip (SoC) **102**, a memory **104**, and one or more I/O devices **106(0)-106(N)**. The system **100** is operable to minimize unnecessary dirty data transfers among two or more cores **108(0)-108(N)** of a processor **110**. Each of the cores **108(0)-108(N)** is associated with a corresponding cache memory **112(0)-112(N)** that is configured to store at least a portion of data **114** obtained from the memory

104 for use by its respective core **108**. The illustrated SoC **102** also includes a DMA controller **116**, a memory controller **118**, and a cache coherence controller **120** which will be described in detail below. The DMA controller **116**, the memory controller **118**, and the cache coherence controller **120**, in one or more implementations, include hardware circuitry designed to enable the corresponding controllers to perform various operations described herein.

(34) Examples of devices or apparatuses in which the system **100** is implemented include, but are not limited to, a personal computer (e.g., a desktop or tower computer), a smartphone or other wireless phone, a tablet or phablet computer, a notebook computer, a laptop computer, a wearable device (e.g., a smartwatch, an augmented reality headset or device, a virtual reality headset or device), an entertainment device (e.g., a gaming console, a portable gaming device, a streaming media player, a digital video recorder, a music or other audio playback device, a television, a set-top box), an Internet of Things (IoT) device, an automotive computer, and other computing devices or systems.

(35) The processor **110** is an electronic circuit that performs various operations on and/or using the data **114** stored in the memory **104**. In the illustrated example, the processor **110** is combined with the DMA controller **116**, the memory controller **118**, and the cache coherence controller **120** as part of the SoC **102**. In one or more implementations, additional or alternative components are included as part of the SoC **102**. Examples of the processor **110** include, but are not limited to, a central processing unit (CPU), a graphics processing unit (GPU), a field programmable gate array (FPGA), an accelerated processing unit (APU), and a digital signal processor (DSP), to name a few. Moreover, the SoC **102** itself, in some implementations, is a CPU, a GPU, an FPGA, an APU, or a DSP, for example.

(36) In the illustrated example, the processor **110**, the DMA controller **116**, the memory controller **118**, and the cache coherence controller **120** are coupled to one another via one or more wired or wireless connections. Example wired connections include, but are not limited to, memory channels, buses (e.g., data, system, I/O, and/or memory busses), interconnects, through silicon vias, traces, and planes. In one example, at least one of the connections is a Peripheral Component Interconnect Express (PCIe) bus. Other example connections include optical connections, fiber optic connections, and/or connections or links based on quantum entanglement.

(37) As introduced above, the processor **110** includes the cores **108**, which are individual processing units that read and execute instructions (e.g., of a program). Example instructions include instructions to add the data **114**, to move the data **114** (e.g., from the memory **104** to the cache **112**), and to branch the data **114**. Aspects of the techniques described herein address dirty data transfers among the two or more caches **112**. Accordingly, in the non-limiting example described herein, the processor **110** is a multi-core processor having more than one core **108**.

(38) The processor **110** is capable of being implemented in accordance with various processor architectures. Some example architectures include, but are not limited to, a monolithic processor architecture, a chiplet-based processor architecture (e.g., core complex “CCX” and core complex die “CCD”), multi-chip modular processor architecture (e.g., multiple independent chips mounted on a single package), and a “big. LITTLE” processor architecture (e.g., a number of “big” high-performance cores along with a number of “LITTLE” power-efficient cores on the same processor die). Moreover, the processor **110** is capable of being implemented on the same die, the same package, or across multiple connected dies or packages. The processor **110**, in one or more implementations, is part of an SoC (e.g., as shown in the SoC **102**) or a system-in-package (SiP) along with other components illustrated and described with reference to FIG. 1.

(39) The cache **112**, in some implementations, is or includes a level 1 cache (L1 cache) that stores data (e.g., at least a portion of the data **114**) and instructions that are immediately needed by the respective core **108** without any significant delay. The cache **112**, in some implementations, also is or includes a level 2 cache (L2 cache). The L2 cache alternatively is part of a shared cache (not shown) that is shared among more than one core **108**. A shared cache, in some implementations, is

or includes a level 3 cache (L3 cache). A core **108** first checks its respective cache **112** (e.g., L1 and L2) for the data **114** needed to perform operations before then moving on to another cache level (e.g., a shared L3 cache), and then the memory **104** as part of a memory hierarchy.

(40) The memory **104** is a device or system that is used to store information, such as the data **114**, for immediate use in a device, such as by the processor **110** or a core **108** thereof, by a DMA engine **122** of the I/O device **106**, or by an in-memory processor (not shown), which is referred to as a processing-in-memory (PIM) component. The memory **104** is implemented as a printed circuit board, on which one or more physical memory chips are disposed (e.g., via physical and communicative coupling using one or more sockets). In other words, the memory **104** is mounted on a printed circuit board and this construction, along with the communicative couplings (e.g., control signals and buses) and one or more sockets integral with the printed circuit board, form the memory **104**.

(41) In at least one example, the memory **104** corresponds to or includes volatile memory, examples of which include random-access memory (RAM), dynamic random-access memory (DRAM), synchronous dynamic random-access memory (SDRAM), static random-access memory (SRAM), and memristors. The memory **104** is packaged or configured in any of a variety of different manners. Examples of such packaging or configuring include a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), a registered DIMM (RDIMM), a non-volatile DIMM (NVDIMM), a ball grid array (BGA) memory permanently attached to (e.g., soldered to) a printed circuit board, and so forth.

(42) Examples of types of DIMMs include, but are not limited to, synchronous dynamic random-access memory (SDRAM), double data rate (DDR) SDRAM, double data rate 2 (DDR2) SDRAM, double data rate 3 (DDR3) SDRAM, double data rate 4 (DDR4) SDRAM, and double data rate 5 (DDR5) SDRAM. In at least one variation, the memory **104** is configured as or includes a SO-DIMM or an RDIMM according to one of the above-mentioned standards (i.e., DDR, DDR2, DDR3, DDR4, and DDR5).

(43) Alternatively, or in addition, the memory **104** corresponds to or includes non-volatile memory, examples of which include flash memory, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electronically erasable programmable read-only memory (EEPROM), and non-volatile random-access memory (NVRAM), such as phase-change memory (PCM) and magneto resistive random-access memory (MRAM). The memory **104** is configurable in a variety of ways capable of supporting requests from the I/O device **106** for accessing the memory **104** to read data **114** from and/or write data **114** to the memory **104**.

(44) Further examples of memory configurations include low-power double data rate (LPDDR), also known as LPDDR SDRAM, which is a type of synchronous dynamic random-access memory. In variations, LPDDR consumes less power than other types of memory and/or has a form factor suitable for mobile computers and devices, such as mobile phones. Examples of LPDDR include, but are not limited to, low-power double data rate 2 (LPDDR2), low-power double data rate 3 (LPDDR3), low-power double data rate 4 (LPDDR4), and low-power double data rate 5 (LPDDR5). It is to be appreciated that the memory **104** is configurable in a variety of ways without departing from the spirit or scope of the described techniques.

(45) In one or more implementations, a device driver **124** issues an I/O command **126** to the I/O device **106**, such as to control the I/O device **106**. For example, the device driver **124** issues the I/O command **126** in connection with execution of an application (not shown) on the processor **110**. Broadly, the I/O command **126** instructs the I/O device **106** to perform an I/O operation, such as by accessing one or more portions of the memory **104**. In one or more implementations, the I/O command **126** includes a size of the I/O operation and a location of a respective DMA buffer **128** in the memory **104**. The device driver **124** is an instantiation of software executing on the processor **110** that enables communication between an operating system (e.g., implemented by the system

100) and the I/O device **106**. Broadly, the device driver **124** acts as an intermediary between external hardware (i.e., the I/O device **106**) and the operating system.

(46) The I/O device **106** is any of a variety of hardware devices capable of handling an I/O command **126** from a respective device driver **124** executing on the processor **110** and carrying out the corresponding I/O operation by accessing (e.g., directly) one or more portions of the memory **104**. Examples of such I/O devices include but are not limited to data stores, such as one or more of a solid-state drive (SSD), a flash drive, a hard disk drive (HDD), a universal serial bus (USB) drive, and so on; communication devices, such as one or more of a network interface adapter (NIC), a Bluetooth® (or other type of wireless communication) adapter and/or dongle, a modem, and so on; and/or peripherals such as printers, monitors, and input devices (e.g., mouse and keyboard). It is to be appreciated that in variations, the I/O device **106** is a different device than those described without departing from the spirit or the scope of the techniques described herein.

(47) In accordance with the described techniques, the I/O device **106** sends requests **129** (e.g., multiple memory access requests) to the DMA controller **116** to access (e.g., read from or write to) various portions of the memory **104** to carry out the I/O operation, such as responsive to the I/O command **126**. In at least one variation, for instance, the DMA engine **122** generates the requests **129** and causes the requests **129** to be transmitted over a connection provided by the DMA controller **116** for directly accessing respective portions of the memory **104** (e.g., isolated as the DMA buffers **128(0)**-**128(N)** corresponding to the DMA engines **122(0)**-**122(N)**), in accordance with DMA techniques.

(48) In one or more implementations, the DMA engine **122** initiates transfers of the data **114** automatically based on preconfigured settings, and/or the DMA engine **122** is programmed by the processor **110** to perform specific transfers. Broadly, the DMA engine **122** enables high-speed data transfers without involving the processor **110**. Rather than relying on the processor **110** to manage data transfers, the DMA engine **122** directly accesses the memory **104** and transfers the data **114** for the I/O device **106**, which improves performance (e.g., throughput of a CPU) and reduces processor (e.g., CPU) overhead.

(49) The DMA engine **122** controls memory access (or at least a portion of the memory access) by the I/O device **106**, such as memory accesses requested in connection with carrying out an I/O operation as instructed by the I/O command **126**. In at least one implementation, the DMA engine **122** is hardware integral with or integrated into the I/O device **106**. In one or more implementations, for example, the DMA engine **122** includes a controller that manages the transfer of the data **114** (or a portion thereof) between the I/O device **106** (e.g., solid state or flash drives of the I/O device **106**) and the memory **104** and further includes a set of buffers for at least temporarily storing the data **114** during the transfer.

(50) The DMA controller **116** is a hardware component (e.g., a special-purpose digital circuit) enabling peripherals, such as the I/O devices **106**, to directly communicate with the memory **104**, bypassing the processor **110** to optimize high-speed data transfers between the I/O devices **106** and the memory **104**. The DMA controller **116** utilizes one or more DMA channels to handle data transfers. Each DMA channel is capable of operating independently of the other DMA channels to enable separate data transfers. The DMA controller **116** operates in various modes, such as burst mode, cycle stealing, scatter-gather, and chain transfer. The DMA controller **116** utilizes specific registers to determine memory addresses and track transfer counts. The DMA controller **116** uses signals like DMA Request (DREQ) and DMA Acknowledge (DACK) for handshaking and arbitrates for control of the memory bus (not shown), potentially locking out the processor **110** during its operations.

(51) The memory controller **118** is a hardware component (e.g., a special-purpose digital circuit) that manages flow of the data **114** between the processor **110** and the memory **104**. The memory controller **118** handles requests from the core(s) **108** and/or other devices (e.g., peripheral devices) to read from and/or write to the memory **104**. Although depicted in the example system **100** as

being implemented as part of the SoC **102**, in some alternative implementations, the memory controller **118** is implemented separately from the SoC **102**, such as part of the memory **104**. As another alternative, the memory controller **118** is implemented as part of the processor **110**.

(52) In accordance with one or more implementations, the memory controller **118** is associated with a single channel of the memory **104**. For instance, the system **100** is configured to include a plurality of different memory controllers **118** on the same or different SoCs **102**, one for each of a plurality of channels of the memory **104**. The techniques described herein are thus performable using a plurality of different memory controllers **118** to schedule requests for different channels of the memory **104**. In some implementations, a single channel in the memory **104** is allocated into multiple pseudo-channels. In such implementations, the memory controller **118** is configured to schedule requests for different pseudo-channels of a single channel in the memory **104**.

(53) The cache coherence controller **120** is a hardware component (e.g., a special-purpose digital circuit) configured to execute instructions, such as instructions to implement an adaptive algorithm **130**, to ensure consistency of the data **114** across the caches **112** of the processor **110**. In addition, the cache coherence controller **120** ensures coherence between the caches **112** and the DMA buffers **128** in the memory **104**. During a write request (e.g., of the requests **129**) to the memory **104**, the cache coherence controller **120** checks if the contents of the destination memory location is stored in one or more of the caches **112**. If so, to maintain data integrity of the system **100**, the cache coherence controller **120** either invalidates the cache entry/entries or updates the cache entry/entries (e.g., changing the previous “dirty data” to “clean data”). During a read request (e.g., of the requests **129**), the cache coherence controller **120** ensures that any modified data in the cache(s) **112** is appropriately updated before a DMA operation reads it. By performing these checks and actions, the cache coherence controller **120** ensures that operations of both the processor **110** and the DMA controller **116** are accessing the most up-to-date version of the data **114**.

(54) FIG. 2 is a block diagram depicting a non-limiting example of the cache coherence controller **120** and the cache **112**. In particular, the illustrated example depicts two caches **112(0)**, **112(1)** controlled by the cache coherence controller **120**. Each of the caches **112** includes one or more cache lines **202**. The cache lines **202** represent the fundamental unit of data transfer between the memory **104** and the caches **112**. Instead of fetching individual bytes or words from the memory **104**, the caches **112** use the cache lines **202** to exploit spatial locality and improve efficiency of the system **100**. The cache lines **202** typically store between 32 bytes and 256 bytes of data **114**.

(55) The data **114** stored in the cache lines **202** is described herein as being “clean” or “dirty.” In the illustrated example, the cache_0 **112(0)** includes one or more cache lines **202(0)** that contain “clean” data **204**, and the cache_1 **112(1)** includes one or more cache lines **202(1)** that contain “dirty” data **206**. The data **114** is considered “dirty” (i.e., “dirty” data **206**) when it has been modified in the cache **112** and is different from the corresponding data **114** in the memory **104**. Conversely, the data **114** is considered “clean” (i.e., “clean” data **204**) when it has not been modified in the cache **112** since it was fetched from the memory **104** or when it matches the corresponding data **114** in the memory **104**.

(56) In the illustrated example, the caches **112** also include metadata **208** expressed in the form of status bits to indicate the state of the cache lines **202**. The illustrated example shows four possible states for the cache lines **202**, including a modified state **210**, an exclusive state **212**, a shared state **214**, and an invalid state **216**. In the modified state **210**, the cache **112** has a version of the cache line **202** that has been modified (i.e., different data **114** than the memory **104**). Cache lines **202** in the modified state **210** contain “dirty” data **206**. In the exclusive state **212**, the cache **112** has an exclusive copy of the cache line **202** (i.e., no other cache **112** has it). Cache lines **202** in the exclusive state **212** contain “clean” data **204** because the data **114** matches the memory **104**. In the shared state **214**, the cache line **202** is stored in one or more other caches **112**. The data **114** in the shared state **214** also matches the memory **104**. Cache lines **202** in the shared state **214** contain “clean” data **204** because coherency exists among the caches **112** and the memory **104**. In the

invalid state, the cache line **202** is not valid, and therefore the data **114** contained in the cache line **202** is not reliable and considered to be “dirty” data **206**.

(57) Although the illustrated metadata **208** is shown having four possible states, additional states can be implemented without departing from the spirit or scope of the described techniques. For example, in some implementations, an owned state is used to describe that the cache **112** has a modified copy of the cache line **202** and that the cache **112** is responsible for updating the memory **104**.

(58) The cache coherence controller **120** includes hardware circuitry **218** configured to execute the adaptive algorithm **130**. In one or more implementations, the hardware circuitry **218** is a digital circuit configured to implement the adaptive algorithm **130** in hardware to perform various operations. In other implementations, the hardware circuitry **218** is configured to execute instructions of the adaptive algorithm **130** to perform various operations. The cache coherence controller **120** also includes a controller memory **219**. The controller memory **219** is implemented as part of the hardware circuitry **218** or separate from the hardware circuitry **218**. The controller memory **219** is configured similar to or the same as the memory **104** but as dedicated memory for the cache coherence controller **120**. In one or more implementations, the controller memory **219** is a portion of the memory **104** that is dedicated for use by the cache coherence controller **120**.

(59) The controller memory **219** which is configured to store a cache line history **220** having dirty data transfer records **222**. The cache line history **220**, in some implementations, is stored in memory as part of the hardware circuitry **218** or separate from the hardware circuitry **218**. The cache coherence controller **120** tracks the transfer of “dirty” data **206** among the caches **112**. For example, if the “dirty” data **206** shown in the cache_1 **112(1)** is transferred to the cache_0 **112(0)**, the cache coherence controller **120** updates the dirty data transfer records **222** to reflect this dirty data transfer. The dirty data transfer records **222** include, for example, the source cache (e.g., the cache_1 **112(1)**) and the destination cache (e.g., the cache_0 **112(0)**). The dirty data transfer records **222** also include, for example, a memory address associated with the data **114** when the data **114** was originally transferred from the memory **104** to the cache_1 **112(1)** (e.g., before the data **114** was modified to become the “dirty” data **206**).

(60) In one or more implementations, when a dirty data transfer occurs, the cache coherence controller **120** also increments a dirty data transfer counter **224** (e.g., plus one). A single dirty data transfer counter **224** is depicted and counts dirty data transfers for all of the caches **112**. Multiple dirty data transfer counters **224** are also contemplated, such as one dirty data transfer counter **224** associated with each one of the caches **112**.

(61) FIG. 3 is a block diagram depicting non-limiting example interactions **300** among various components of the system **100** depicted in FIG. 1. In particular, the illustrated example includes the memory **104** having the DMA buffer **128** storing the data **114**, the I/O device **106** having the DMA engine **122**, the core **108** associated with the cache **112**, the DMA controller **116**, and the cache coherence controller **120** having the adaptive algorithm **130**.

(62) The DMA engine **122** generates a DMA request **302** (e.g., as one of the requests **129**) and sends the DMA request **302** to the DMA controller **116**. The DMA controller **116** determines if the DMA request **302** is a read request or a write request. For a write request, the DMA request **302** includes data **114** to be written to the memory **104**, and in particular, to the DMA buffer **128** assigned to the DMA engine **122** of the I/O device **106**. The DMA controller **116** receives the write request and sends a write command **304** to the memory **104**. For a read request, the DMA request **302** identifies the DMA buffer **128** from which to read the data **114**. For example, the DMA engine **122** identifies the DMA buffer **128** based on the I/O command **126** received from the processor **110** (shown in FIG. 1). The DMA controller **116** receives the read request and requests the data **114** from the memory **104** in a read command **306**.

(63) In response to the DMA request **302**, the DMA controller **116** also determines whether the core **108** commands the memory bus (i.e., whether the core **108** is actively using the memory bus to

communicate with the memory **104**). If so, the DMA controller **116** generates a stop command **308** directed to the core **108**. The stop command **308** instructs the core **108** to give up control of the memory bus to allow the DMA controller **116** access to the memory **104** to perform the write command **304** or the read command **306** on behalf of the DMA engine **122** of the I/O device **106**. When the DMA controller **116** completes operations associated with the DMA request **302**, the DMA controller **116** generates a DMA response **310** and sends the DMA response **310** to the DMA engine **122**. The DMA response **310** includes the data **114** retrieved responsive to the read command **306** or an acknowledgment of the write command **308**. The DMA controller **116** also issues a resume command **312** to the core **108**. The core **108**, in response, resumes control of the memory bus.

(64) Also, responsive to the DMA request **302**, the cache coherence controller **120** executes the adaptive algorithm **130** to determine the cache line **202** that holds the data **114** associated with the DMA request **302**. The cache coherence controller **120** then checks the cache line history **220** to determine if the cache line **202** is associated with a dirty data transfer. If the cache coherence controller **120** determines that the cache line **202** is not associated with a dirty data transfer, the cache coherence controller **120** enters mode A **314** and performs normal operations. In other words, the cache coherence controller **120** does not change the state of the cache line **202**. If, instead, the cache coherence controller **120** determines that the cache line **202** is associated with a dirty data transfer, the cache coherence controller **120** enters mode B **316** and executes the adaptive algorithm **130** to generate one or more probes **318** directed to the cache **112**. The probes **318** either invalidate the cache line **202** or convert the data **114** in the cache line **202** from dirty to clean based upon the state of the cache line **202**.

(65) FIG. **4** is a flow diagram depicting a procedure **400** in a non-limiting example implementation of the DMA controller **116** configured to perform operations to minimize I/O dirty data transfers. The procedure **400** begins and the I/O device **106** issues the DMA request **302** to write data **114** to or read data **114** from the memory **104** (block **402**). For example, the DMA engine **122** of the I/O device **106** generates the DMA request **302** and sends the DMA request **302** to the DMA controller **116**. In at least one implementation, the DMA engine **122** is hardware integral with or integrated into the I/O device **106**. In one or more implementations, for example, the DMA engine **122** includes a controller that manages the transfer of the data **114** (or a portion thereof) between the I/O device **106** (e.g., solid state or flash drives of the I/O device **106**) and the memory **104** and further includes a set of buffers for at least temporarily storing the data **114** during the transfer.

(66) Responsive to the DMA request **302**, the DMA controller **116** checks the memory bus status (block **404**) to determine whether the core **108** is actively using the memory bus (block **406**). For example, the DMA controller **116** checks the memory bus status using bus request and grant signals, and employs arbitration logic and priority scheduling to ensure smooth and conflict-free access to the memory **104** to write or read the data **114**. After the data **114** is transferred, the DMA controller **116** releases the memory bus and continues monitoring the status of the memory bus to prevent conflicts with the core **108** and/or other devices.

(67) If the processor **110** (e.g., the core **108**) is actively using the memory bus, the DMA controller **116** issues the stop command **308** to the core **108** (block **408**) and the DMA controller **116** gains control of the memory bus (block **410**). The stop command **308** instructs the core **108** to give up control of the memory bus to allow the DMA controller **116** access to the memory **104** to perform the write command **304** or the read command **306** on behalf of the DMA engine **122** of the I/O device **106**. The DMA controller **116** then reads the data **114** from the memory **104** and provides the data **114** to the I/O device **106** (block **412**). Alternatively, the DMA controller **116** writes the data **114** to the memory **104** on behalf of the I/O device **106** (block **412**). In this manner, the DMA controller **116** bypasses the processor **110** for reading the data **114** from or writing the data **114** to the memory **104**.

(68) Returning to block **406**, if the processor **110** (e.g., the core **108**) is not actively using the

memory bus, the DMA controller **116** reads the data **114** from the memory **104** and provides the data **114** to the I/O device **106** (block **412**). Alternatively, the DMA controller **116** writes the data **114** to the memory **104** on behalf of the I/O device **106** (block **412**).

(69) FIG. 5 is a flow diagram depicting a procedure **500** in a non-limiting example implementation of the cache coherence controller **120** configured to perform operations to selectively execute the adaptive algorithm **130**. The procedure **500** begins and the cache coherence controller **120** detects the DMA request **302** (block **502**). For example, when the I/O device **106** requires data transfer, the I/O device **106** (e.g., via the DMA engine **122**) sends a signal to the DMA controller **120**. The DMA controller **120** detects the DMA request **302**, which then initiates the DMA process to transfer data directly between the I/O device **106** and the memory **104**, bypassing the processor **110**.

(70) The cache coherence controller **120** identifies the cache line **202** associated with the DMA request **302** (block **504**). That is, the cache coherence controller **120** identifies the cache line **202**, if any, that holds the requested data **114**). The cache coherence controller **120** then checks the cache line history **220** for the cache line **202** (block **506**) to determine whether the cache line history **220** contains a dirty transfer record **222** for the cache line **202** (block **508**). The dirty data transfer records **222** include, for example, the source cache (e.g., the cache_1 **112(1)**) and the destination cache (e.g., the cache_0 **112(0)**). The dirty data transfer records **222** also include, for example, a memory address associated with the data **114** when the data **114** was originally transferred from the memory **104** to the cache_1 **112(1)** (e.g., before the data **114** was modified to become the “dirty” data **206**).

(71) If the cache line history **220** does not contain a dirty data transfer record **222** for the cache line **202** (as determined at block **508**), then the cache coherence controller **120** performs normal operation in mode A **314**. The normal operation in mode A **314** does not change the cache line state. If, however, the cache line history **220** contains a dirty data transfer record **222** for the cache line **202** (as determined at block **508**), the cache coherence controller **120** executes the adaptive algorithm **130** in the mode B **316**. Additional details in this regard are described below with reference to FIG. 6.

(72) FIG. 6 is a flow diagram depicting another procedure **600** in a non-limiting example implementation of the cache coherence controller **120** configured to perform operations to minimize I/O dirty data transfers. The procedure **600** begins and the cache coherence controller **120** determines if the state of the cache line **202** is the exclusive state **212** (block **602**). In the exclusive state **212**, the cache **112** has an exclusive copy of the cache line **202** (i.e., no other cache **112** has it). Cache lines **202** in the exclusive state **212** contain “clean” data **204** because the data **114** matches the memory **104**. If the cache coherence controller **120** determines that the state of the cache line **202** is the exclusive state **212** (as determined at block **602**), the cache coherence controller **120** sends the probe **318** to invalidate the cache line **202** (block **604**). In particular, the cache coherence controller **120** changes the state of the cache line **202** from the exclusive state **212** to the invalid state **216** (block **604**). In the invalid state, the cache line **202** is not valid, and therefore the data **114** contained in the cache line **202** is not reliable and considered to be “dirty” data **206**.

(73) If the cache coherence controller **120** instead determines that the state of the cache line **202** is not in the exclusive state **212** (as determined at block **602**), the cache coherence controller **120** determines if the state of the cache line **202** is in the shared state **214** (block **608**). In the shared state **214**, the cache line **202** is stored in one or more other caches **112**. The data **114** in the shared state **214** also matches the memory **104**. Cache lines **202** in the shared state **214** contain “clean” data **204** because coherency exists among the caches **112** and the memory **104**. The data **114** is considered “clean” (i.e., “clean” data **204**) when it has not been modified in the cache **112** since it was fetched from the memory **104** or when it matches the corresponding data **114** in the memory **104**.

(74) If the cache coherence controller **120** determines that the cache line **202** is in the shared state

214 (as determined at block **608**), the cache coherence controller **120** sends the probe **318** to convert the cache line **202** from a dirty shared state to a clean shared state (block **610**).

(75) The cache coherence controller **120** then returns to normal operation (block **606**). If the cache coherence controller **120** instead determines that the cache line **202** is not in the shared state **214** (block **608**), the cache coherence controller **120** then returns to normal operation (block **606**). For example, the normal operation is mode A **314** in which the cache line state does not change.

(76) It will be understood that many variations are possible based on the disclosure herein. Although features and controls are described above in particular combinations, each feature or control is usable alone without the other features and controls or in various combinations with or without other features and controls.

(77) The various functional units illustrated in the figures and/or described herein (including, where appropriate, the system **100**, the SoC **102**, the memory **104**, the I/O device **106**, the cores **108**, the processor **110**, the caches **112**, the DMA controller **116**, the memory controller **118**, the cache coherence controller **120**, the DMA engines **122**, and the device driver **124**) are implemented in any of a variety of different manners such as hardware circuitry, software or firmware executing on a programmable processor, or any combination of two or more of hardware, software, and firmware. The methods provided are implemented in any of a variety of devices, such as a general-purpose computer, a processor, or a processor core. Suitable processors include, by way of example, a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a graphics processing unit (GPU), a parallel accelerated processor, a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs) circuits, any other type of integrated circuit (IC), and/or a state machine.

(78) In one or more implementations, the methods and procedures provided herein are implemented in a computer program, software, or firmware incorporated in a non-transitory computer-readable storage medium for execution by a general-purpose computer or a processor. Examples of non-transitory computer-readable storage mediums include a read only memory (ROM), a random-access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs).

Claims

1. A system comprising: a processor; a memory configured to store data; a cache configured to store a portion of the data stored in the memory for execution by the processor; and a cache coherence controller comprising a cache line history, the cache coherence controller configured to: identify a cache line associated with a direct memory access request, the direct memory access request associated with an operation involving the data; and responsive to the cache line history including a dirty data transfer record corresponding to the cache line, selectively send a probe to the cache, the dirty data transfer record including information that indicates whether a dirty transfer was triggered by a previous request for the cache line.
2. The system of claim 1, wherein the cache coherence controller is configured to selectively send the probe to the cache based on a state of the cache line, the state of the cache line comprising a modified state, an exclusive state, a shared state, or an invalid state.
3. The system of claim 2, wherein the cache coherence controller is further configured to, in response to determining that the state of the cache line is the exclusive state, send the probe to the cache to invalidate the cache line.
4. The system of claim 2, wherein the cache coherence controller is further configured to, in response to determining that the state of the cache line is the shared state, send the probe to convert the state of the cache line from a dirty shared state to a clean shared state.

5. The system of claim 1, wherein the processor comprises a first core and a second core, and wherein the cache comprises a first cache corresponding to the first core and a second cache corresponding to the second core.
6. The system of claim 5, wherein the first cache stores the portion of the data in the cache line.
7. The system of claim 5, wherein the first core transfers the portion of the data from the first cache to the second cache.
8. The system of claim 5, wherein the first core modifies the portion of the data to create a new portion of the data that is different from the portion of the data.
9. The system of claim 8, wherein the cache coherence controller is further configured to update the cache line history to reflect that the portion of the data was modified before being transferred to the second cache.
10. The system of claim 1, wherein the operation involving the data comprises an input or output operation.
11. The system of claim 1, wherein the cache coherence controller is configured to selectively send the probe to the cache by sending the probe to the cache if the information in the dirty data transfer record indicates that the dirty transfer was triggered by the previous request for the cache line.
12. The system of claim 1, wherein the cache coherence controller is configured to selectively send the probe to the cache by not sending the probe to the cache if the information in the dirty data transfer record indicates that the dirty transfer was not triggered by the previous request for the cache line.
13. A cache coherence controller comprising: a memory configured to store a cache line history; and a hardware circuitry configured to: identify a cache line associated with a direct memory access request, the direct memory access request associated with an operation involving data stored in a cache; and responsive to the cache line history including a dirty data transfer record corresponding to the cache line, selectively send a probe to the cache, the dirty data transfer record including information that identifies whether a dirty transfer was triggered by a previous request for the cache line.
14. The cache coherence controller of claim 13, wherein hardware circuitry comprises the memory.
15. The cache coherence controller of claim 13, wherein the hardware circuitry is configured to selectively send the probe to the cache based on a state, the state comprising a modified state, an exclusive state, a shared state, or an invalid state.
16. The cache coherence controller of claim 15, wherein the hardware circuitry is further configured to, in response to determining that the state of the cache line is the shared state, send the probe to convert the state of the cache line from a dirty shared state to a clean shared state.
17. The cache coherence controller of claim 13, wherein the hardware circuitry is further configured to update the cache line history to reflect that a portion of the data was modified before being transferred to the cache.
18. The cache coherence controller of claim 13, wherein the operation involving the data comprises an input operation or an output operation.
19. A method comprising: detecting, by a cache coherence controller, a direct memory access request from a direct memory access engine of an input/output device, the direct memory access request associated with an input/output operation performed by the input/output device; and responsive to detecting the direct memory access request, selectively executing, by the cache coherence controller, an adaptive algorithm to ensure coherency of data between a plurality of caches and a memory, the data associated with the input/output operation, wherein the cache coherence controller selectively executes the adaptive algorithm based on a dirty data transfer record that includes information regarding previous requests to the caches.
20. The method of claim 19, wherein selectively executing, by the cache coherence controller, the adaptive algorithm comprises executing, by the cache coherence controller, the adaptive algorithm

in response to the data being associated with a dirty data transfer from a first cache of the plurality of caches to a second cache of the plurality of caches.
